



Gimme feasibility and shit

Short version: your docs say the one-cell OpenBraille servocam MVP is technically buildable this semester if you follow the test register, but the patent story is deliberately conservative and treated as “future hypothesis,” while VibeGuard and TrueMoist carry more schedule/facility risk. ^[1] ^[2] ^[3]

What “feasibility” means in these files

The Concept Evidence Matrix separates three layers: (1) engineering identity and MVP, (2) minimum demonstrable success, and (3) whether Phase-4/“next phase” is actually authorized.

^[1] It explicitly warns that “Ready with Minor Refinements” in an architecture report is not the same as “Phase 4 has started,” and that macro prototypes don’t automatically prove commercial-scale designs or patentability. ^[1] ^[3]

OpenBraille engineering feasibility

OpenBraille’s frozen MVP is a single six-dot cell at macro scale with six SG90-class servos, one eccentric cam per servo, follower-guided vertical pins, an ESP32 using direct PWM, an FDM frame, and an external 5 V $\approx$ 2 A supply plus bulk capacitance. ^[1] ^[3] The matrix estimates a BOM in the 2,060–2,850 range and calls out critical mechanical risks (pin/cam friction, tolerance stack-up, tactile height, wear) and electrical risk (six-servo peaks vs 5 V/2 A rail), but all of these are explicitly targeted by short, pre-registered tests in the Uncertainty and Test Register. ^[1] ^[2]

The OpenBraille uncertainties U-OB-01...07 cover: verifying whether SG90 actually has a usable passive torsion-spring return, proving active cam retraction is reliable if passive return fails, checking six-servo current demands vs the 5 V/2 A supply and capacitors, 3D-printed mechanical tolerances over 1,000 cycles, and designing a blinded tactile-identification protocol with quantitative acceptance criteria. ^[2] Each test has concrete equipment, 1–4 day durations, and “Immediate” or early-semester urgency, which is exactly the profile of “feasible but only if you front-load the bench tests before CAD freeze and assembly.” ^[2] ^[3]

OpenBraille patent feasibility

The pre-build patent-case simulation states that the frozen semester MVP is a “strong semester prototype and a Track D no-current-filing patent position,” because low-cost servo-cam Braille cells and motor-driven cam pin actuation with position sensing are already present in recent US and EP patents and academic work. ^[3] It concludes that the current D11 bundle (calibration, current sensing, retry logic, replacement acceptance, etc.) is still just an aggregation of familiar elements until comparative experiments show a cooperative physical effect that actually lowers silent wrong tactile states beyond strong baselines. ^[3]

The simulation identifies a more promising but speculative future direction—Verified Pin-Datum Cell VPD-C7—where the same mechanically registered datum both sets pin endpoints and produces a pin-side endpoint electrical event, with current used only as independent effort evidence; but it is explicit that this redesign is not disclosed in the frozen architecture and needs new mechanical/electrical design, displacement ground truth, wear/contamination tests, and a refreshed patent search before anyone calls it an invention. ^[3] In other words, engineered OpenBraille MVP: yes; near-term patent: intentionally “not yet,” with feasibility framed as “research hypothesis that only survives if later data beat obvious combinations of HumanWare/Enhanced Vision/current-sensing prior art.” ^[3]

VibeGuard feasibility

VibeGuard’s MVP identity is a passive, single-node edge condition-monitoring system: ADXL345 accelerometer on a rigid mount, ESP32, FFT-based features plus Mahalanobis statistics and spectral subtraction, and a damped motor/fan rig with an eccentric-mass fault. ^[1] The Evidence Matrix and Uncertainty Register both stress that bench-rig normal vs deliberate abnormal classification is on-scope, while dense-floor attribution (one node correctly attributing a fault to its own machine in a crowded environment) is explicitly beyond the minimum demonstrable success and has unresolved site-access and protocol issues. ^{[1] [2]}

Uncertainties U-VG-01...07 require you to (a) prove ADXL345’s bandwidth is enough for all claimed MVP phenomena, (b) show rigid remounts don’t dominate classification, (c) test whether spectral subtraction actually improves discrimination vs simpler thresholds, (d) get real dense-floor ground truth, and (e) demonstrate repeatable eccentric-mass fault simulation and bench classifier accuracy with pre-registered statistical thresholds. ^[2] Many of these have 2–7 day durations but depend on rig build and external site access, which makes VibeGuard more logistically fragile than OpenBraille for a single semester, even though the core bench classification problem is well-scoped. ^{[1] [2]}

TrueMoist feasibility

TrueMoist’s MVP is an embedded soil-moisture instrument that combines genuine capacitive probes, temperature sensing, ADS1115 ADC, local regression-based compensation, and gravimetric ground-truth at $105 \pm 5^\circ\text{C}$ to show that compensated estimates beat raw sensor output on held-out samples under controlled conditions. ^[1] The architecture is labelled as “Ready with Mandatory Early Validation,” with many explicit gates around probe stability, ADC value vs ESP32’s internal ADC, packing/jig effects, oven and precision scale access, controlled salinity treatment, and the throughput needed for roughly 72 independent runs in eight weeks. ^{[1] [2]}

Uncertainties U-TM-01...07 ask whether ADS1115 materially improves noise/drift/RMSE, whether the chosen capacitive probe is monotonic and repeatable, whether a jig can keep packing variance below intended environmental effects, whether institutional oven/scale access is real, and whether a single NaCl treatment produces a stable, non-saturating shift distinct from noise. ^[2] The Concept Evidence Matrix highlights that institutional oven and scale access and the 72-run workflow are direct schedule risks and that a documented 54-run fallback may be needed if throughput is lower, so TrueMoist is feasible only if you lock facilities early and treat schedule planning as a first-week task, not an afterthought. ^{[1] [2]}

What this means for you in practice

Given the way these files are written, OpenBraille is deliberately positioned as the “least risky to build this semester” concept: single-cell macro scale, clear BOM, short bench tests for core mechanical/electrical uncertainties, and a validation plan that you can run with an ESP32 and straightforward firmware, provided you design and pre-register a tactile protocol with the project owner. ^[1] ^[2] ^[3] VibeGuard and TrueMoist are more dependent on access (dense machine floor vs institutional oven/scale) and on bigger statistical datasets, so they’re better treated as “bench-rig and methods prototypes” unless those external conditions are locked in writing. ^[1] ^[2]

If your immediate goal is “get something solid and defensible built and reported,” the feasible path is roughly:

- Front-load the U-OB tests: servo passive/active return, six-servo current adequacy, single-dot mechanical coupon, then design/approve the tactile protocol before committing to full CAD and assembly. ^[2]
- Treat the D11/VPD-C7 patent content strictly as a research roadmap: log all execution data (force, travel, current, failures), but don’t promise patentability until you have comparative evidence against direct sensors and strong prior art. ^[3]
- Only scale VibeGuard beyond a motor-fan bench demo if you can secure a dense-floor site and pre-registered attribution metrics early; otherwise, keep claims to “bench-rig normal vs abnormal classification” with explicit limits. ^[1] ^[2]
- For TrueMoist, do a hard Week-1 facilities check (oven, scale, storage, probes, ADS1115) and be ready to formally adopt the documented 54-run fallback if throughput or access falls short of the ideal 72-run plan. ^[1] ^[2]

If you tell me which of the three you’re actually trying to push right now (OpenBraille vs VibeGuard vs TrueMoist), I can turn this into a very concrete “do this first, then this” checklist keyed directly to the uncertainty IDs and evidence rows in your files. ^[1] ^[2] ^[3]



1. 03_Concept_Evidence_Matrix.md
2. 07_Uncertainty_and_Test_Register.md
3. 17_OpenBraille_PreBuild_Future_Patent_Case_Simulation.md